Analysis of CRISPR-based chromosome labeling with computer vision and machine learning techniques

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Abstract

Spot detection methods for microscopic image analysis provide tools for biological research that uses fluorescent protein labeling to track species of interest. Current detection methods rely on user input of image attributes for analysis, rendering such analyses inefficient and unable to be automated. These methods also exhibit poor accuracy in analysis of low signal-to-noise (SNR) images.

We present a dual approach to improve the accuracy of spot detection: (i) expansion of existing algorithms to autonomously predict image attributes (ii) development of a neural network to perform spot detection in images under low SNR constraint. The automation that our approach introduces to these analyses greatly increases the speed of spot detection in large datasets.

Introduction/Background

Spot detection in microscopic images is often employed to track biological components of a system[1]. Specifically, spot detection is employed to track fluorescent proteins linked to chromosomal loci via CRISPR-based labeling. The analysis of the dynamic proximity of these loci may elucidate insights into the regulation of gene expression[2].

In order to gain insight into this process, many images of the loci must be captured and analyzed. ComDet, an open source plugin for the image processing software ImageJ, is the method of spot detection analysis currently available to the Xing lab. This tool is configured such that significant user interaction is required for each image analysis. Furthermore, the method's accuracy drops significantly in images that have low SNR (SNR<3).

Improvement of the performance and efficiency of spot detection will allow the lab to more quickly and accurately analyze microscopic image data. The ComDet algorithm will be augmented for automation, and a convolutional neural network (CNN) will be developed to improve accuracy of spot detection.

Development of automated spot detection

In order to automate spot detection, methods were created to estimate spot size and SNR values to eliminate required user input.

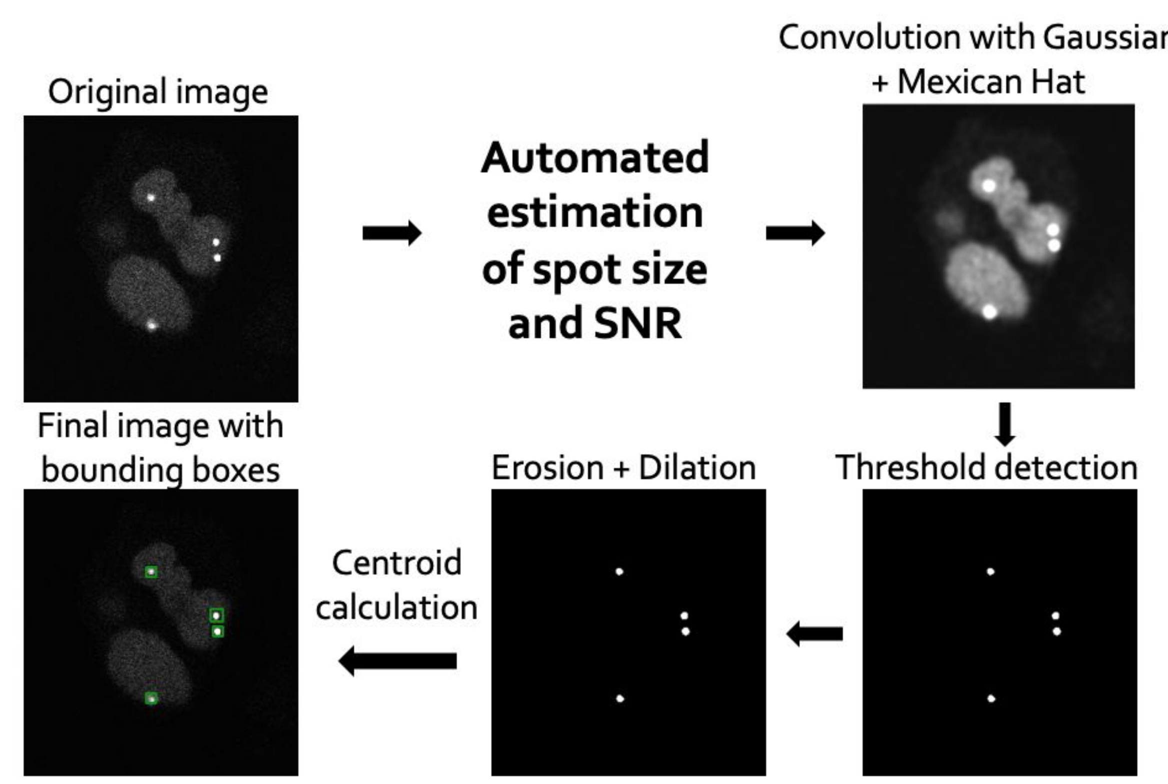

Figure 1. Pipeline of the spot detection method with automated estimation of spot size and SNR

Performance of automated method

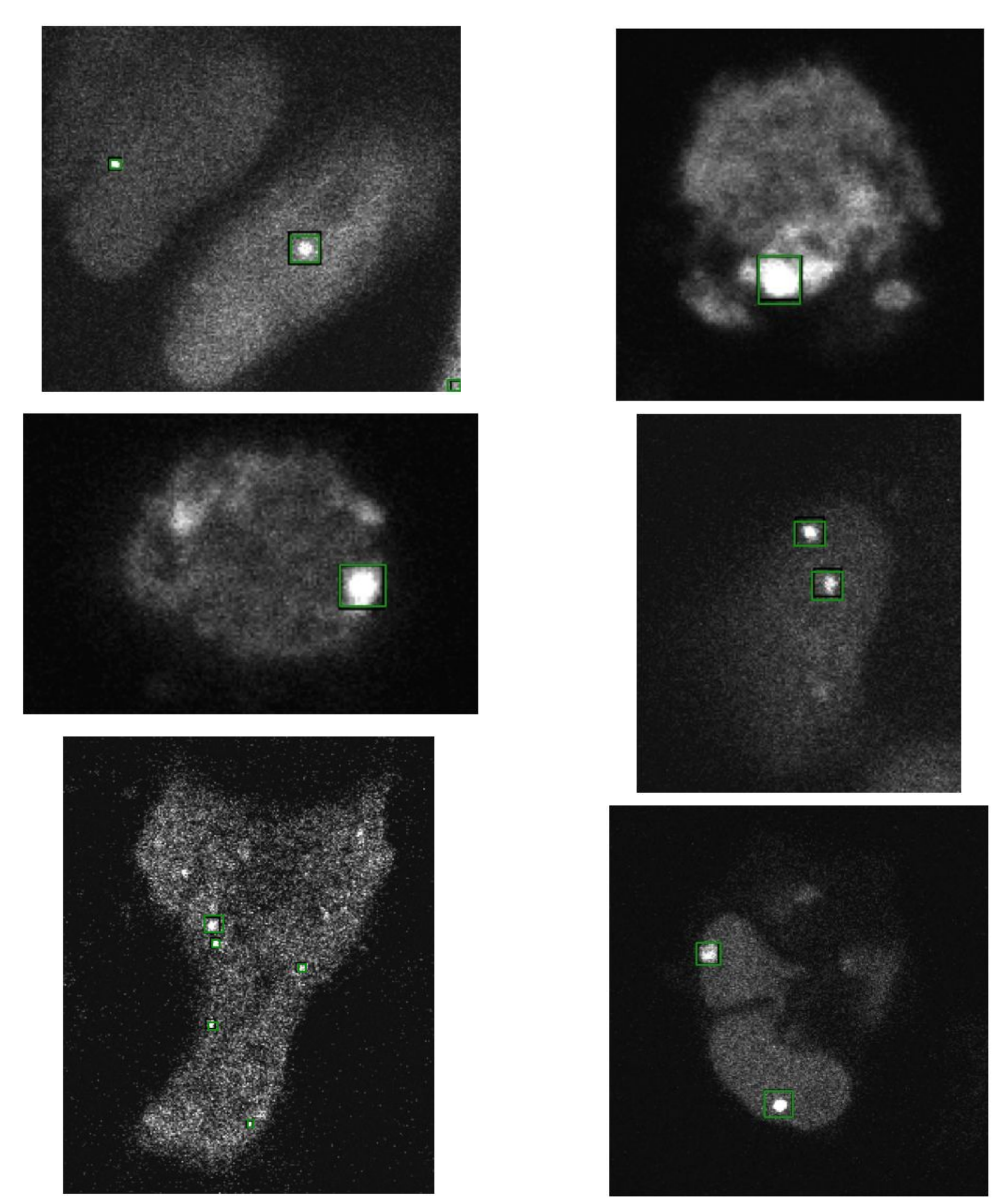

Figure 2. Examples of results from the automated spot detection. Green bounding boxes are laid over original black bounding boxes for visibility.

Development of CNN and training data

A classification CNN architecture was developed (Figure 3). To develop training data, regions of microscopic images that contained bright spots were selected. Trainable Weka Segmentation, an ImageJ plugin, was employed to create labels for the images by producing a binary mask that classified pixels as members of either the background or bright spot (Figure 4).

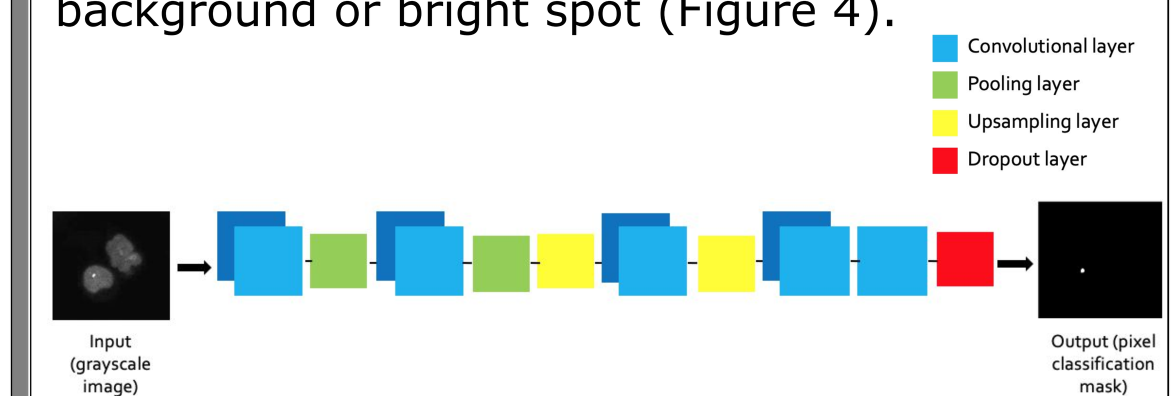

Figure 3. The pixel classification CNN architecture with sample input and output.

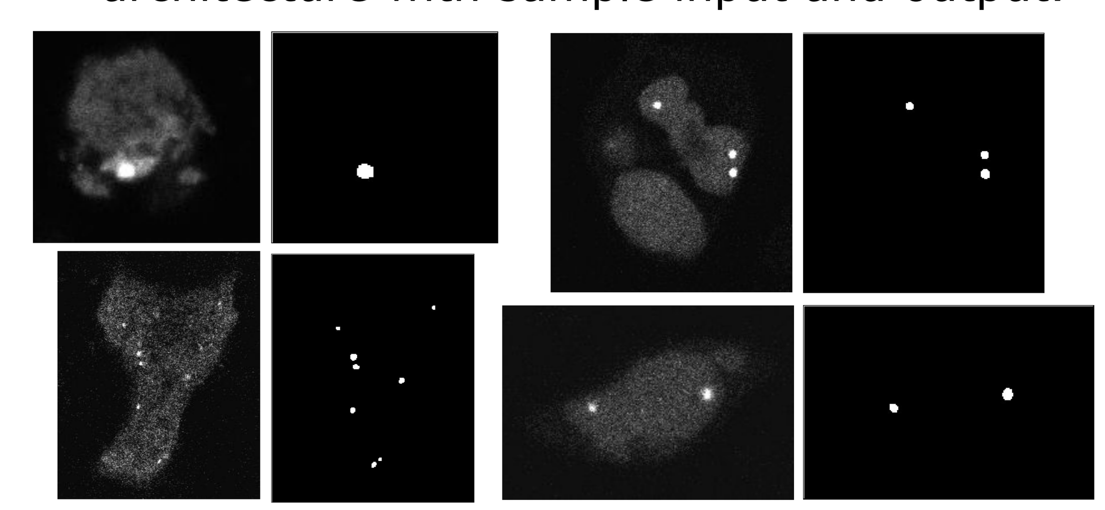

Figure 4. Examples of input images and their corresponding labels for the CNN.

Performance of CNN

With 877 training images and 20 epochs, the CNN reached an accuracy of 89.7% on the 69 test images.

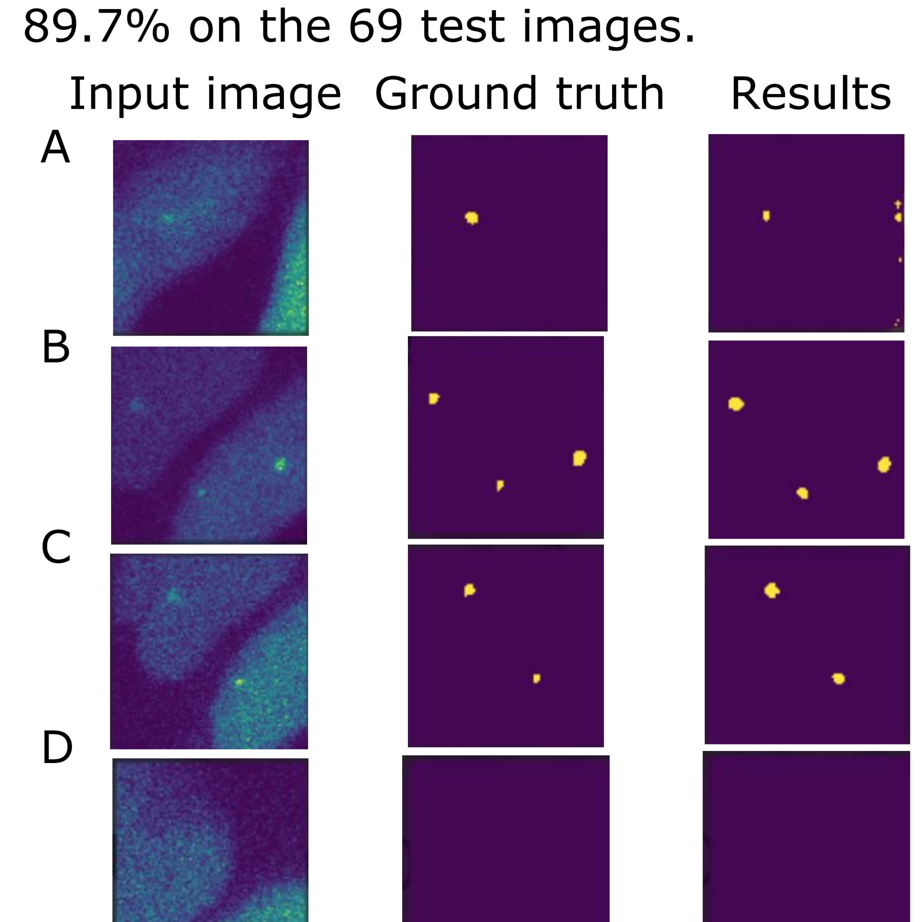

Figure 5. Several examples of results from the CNN spot detection method.

Conclusions

The automated spot detection method appears to perform well across a range of image heterogeneities, but a more thorough performance analysis should be performed on the method to determine the method's most significant weaknesses before improvements can be made. The CNN method performs well on test data (89.7% accuracy), but accuracy should be improved before the method is employed without user supervision.

Future Directions

The accuracy of the automated method will be more rigorously analyzed to determine necessary improvements. The accuracy of the CNN can be further improved by expanding the neural network and training data to be able to analyze images with a wider range of heterogeneities (SNR, spot size, etc.). Spot linking can be employed to develop a full tracking program for image time series.

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References

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